

# Bernard Birendra Das

Data Scientist

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## PROFILE SUMMARY

**Data Scientist and PhD Scholar in Data Science with 2+ years of professional software engineering experience specializing in predictive modeling, statistical inference, and deep learning.** Skilled in PyTorch, TensorFlow, Scikit-learn, and time-series analysis, with research expertise in physiological signals and medical computer vision. Passionate about designing rigorous statistical experiments and production-ready machine learning solutions for complex domain problems.

## WORK EXPERIENCE

**Simplify3x Software Private Limited** • Software Engineer

January 2023 - May 2025 | 2 years 5 months | Bengaluru, Karnataka

- Developed and maintained two impactful modules of SimplifyQA product for the new UI, namely PDF Reports and Pipeline Executions.
- Designed and implemented RESTful APIs and microservices that led to significant improvements in response times and improving overall system scalability.
- Developed dynamic, data-driven dashboards using Angular and D3.js, enabling real-time visualization of key performance metrics for business stakeholders.

## PROJECTS

**Brain Tumor MRI Classification & Segmentation** • TensorFlow, Keras, VGG16, Flask, NumPy, Python

2026

- Preprocessed multi-class MRI scans and fine-tuned a deep convolutional VGG16 backbone, achieving 99.36% test accuracy.
- Built an end-to-end inference pipeline and Flask REST API to deliver instant tumor classification and segmented visualizations.

**ECG Time-Series Predictive Modeling** • PyTorch, Scikit-learn, wfdb, PEFT, Ollama, Python

2026

- Trained 1D ResNet and Random Forest models on MIT-BIH physiological data for automated cardiac anomaly detection.
- Integrated parameter-efficient fine-tuning (PEFT) and local LLM reasoning to explain automated diagnostic predictions.

## EDUCATION

**CHRIST (Deemed-to-be-University), Bangalore**

PhD in Data Science - 3.6/4.0 GPA

Dec 2024 - Present

MCA with Data Analytics - 8.66 CGPA

June 2021 - June 2023

**St. Xavier's College (Autonomous), Kolkata**

BSc (Honours) Computer Science - 7.20 CGPA

June 2018 - June 2021

**Auxilium Convent School, Barasat**

12th Grade, ISC Board - 71.20%

June 2016 - June 2018

10th Grade, ICSE Board - 76.40%

June 2004 - June 2016

## SKILLS

### PROGRAMMING

Languages:

Python • R • SQL • C++ • Java

Machine Learning & DL:

PyTorch • TensorFlow • Scikit-learn • Keras • XGBoost • Hugging Face

Statistical Modeling:

Hypothesis Testing • A/B Testing • PCA •

Time-Series Analysis • Regression

### DATA & PLATFORMS

Data Processing:

Pandas • NumPy • SciPy • Apache Spark • BigQuery

ML Tools & Cloud:

MLflow • Docker • AWS • GCP Vertex AI • Streamlit

### CORE COMPETENCIES

Predictive Modeling • Deep Learning • Feature Engineering • Model Validation • Scientific Research

## CERTIFICATIONS

**Foundational C# with Microsoft**

Developer Certification

Sept 2023

**AWS Academy Graduate**

AWS Academy Cloud Foundations Nov 2022